School of Computer Science Engineering and Technology

Course- B.Tech.	Type- Core
Course Code- **2025CSET106**	Course Name- **Discrete Mathematical Structures**
Year- I	Semester- Even (2^{nd} Semester)

Tutorial- 10

Tutorial No.	Name	CO1	CO2	CO3
10	**Chinese Remainder Theorem, Permutations, and Combinations, Inverse using Euclidean Algorithm**	--	✓	✓

Objective:

- Students will be able to apply the Chinese Remainder Theorem to solve systems of congruences.
- Students will be able to differentiate between permutations and combinations and evaluate the number of possible arrangements and selections.
- Find the multiplicative inverse of an integer modulo $m$using the Euclidean and Extended Euclidean Algorithm.

Questions:

1. Apply Chinese Remainder Theorem (CRT) to solve the following system of congruence equations:

(a) $x \equiv 5 \pmod 7$ and $x \equiv 4 \pmod 9$.
(b) $x \equiv 1 \pmod 2$, $x \equiv 2 \pmod 3$ and $x \equiv 3 \pmod 7$.

2. A child of a number theorist is sorting a large pile of coins (worth less than Rs. 50) into groups of 3 coins each. At the end, the child reports that 2 coins are left over. The child then sorts the coins into groups of 4 and reports that 1 coin is left over. Sorting them into groups of 5, the child again reports that 1 coin is left over.
The number theorist did not originally know how many coins were in the pile, but at this point claims to know the correct number.
Does the number theorist have enough information? If yes, find the number using the Chinese Remainder Theorem.
[Hint: $x \equiv 2 \pmod 3$, $x \equiv 1 \pmod 4$, $x \equiv 1 \pmod 5$]

3. Selection and Committee Formation

From a group of 10 students, answer the following:
(a) In how many ways can 3 students be selected?
(b) In how many ways can a committee of 4 students be formed?
(c) In how many ways can a committee consisting of a president, secretary, and treasurer be formed?
(d) In how many ways can 2 boys and 2 girls be selected if the group contains 6 boys and 4 girls?
(e) In how many ways can a committee of at least one student be formed?

4. Linear and Circular Arrangements
Consider 8 students.
(a) In how many ways can the students be arranged in a row?
(b) In how many ways can 5 out of the 8 students be arranged in a row?
(c) In how many ways can the students be seated around a circular table?
(d) In how many ways can the students be arranged in a row if 3 particular students always sit together?
(e) In how many ways can the students be arranged in a row if two particular students never sit together?

5. Alphabet-Based Permutations and Combinations
Consider the alphabets:
A, B, C, D, E, F, G
(Vowels: A, E Consonants: B, C, D, F, G)
Answer the following:
(a) How many 3-letter words can be formed without repetition from the given alphabets?
(b) How many 4-letter words can be formed with repetition allowed?
(c) How many 3-letter words can be formed without repetition and starting with a vowel?
(d) How many 4-letter words can be formed without repetition, if exactly two letters are vowels?
(e) How many 5-letter words can be formed without repetition, if A and E are always together?

6. Find the multiplicative inverse using the Extended Euclidean Algorithm:
(a) Find the multiplicative inverse of 7 modulo 26 using the Euclidean Algorithm.
(b) Find the multiplicative inverse of 11 modulo 17 using the Euclidean Algorithm.
(c) Find the multiplicative inverse of 17 modulo 43 using the Euclidean Algorithm.